# Task 20

Implement IAM Best Practices

- Create IAM groups: Admin, Dev, ReadOnly
- Apply least privilege policies
- Enable MFA for root
- Create custom policy for S3 restricted folder access

# Step 1: Create IAM Groups

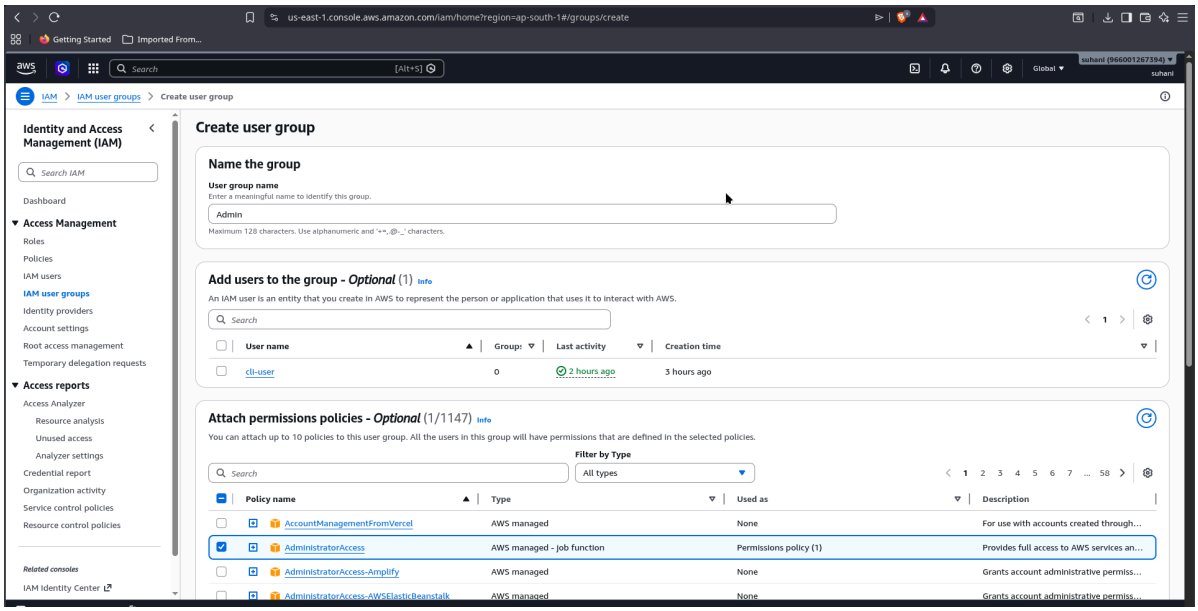
Go to:

- **User groups → Create group**

### Create 3 groups:

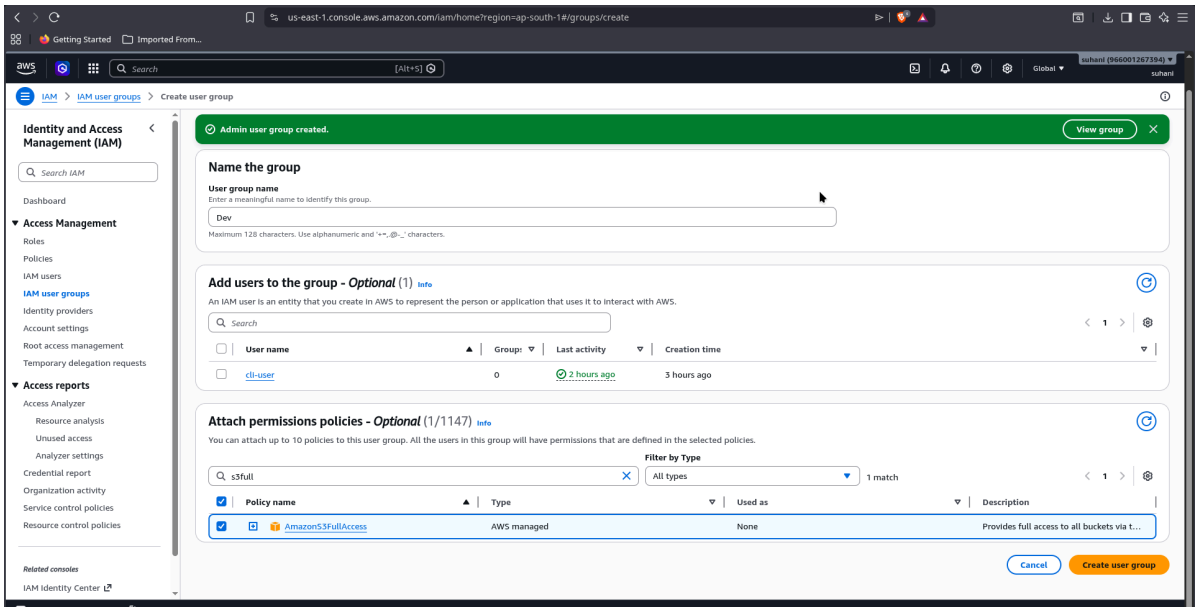
### 1. Admin

- Group name: `Admin`
- Attach policy:
  - ✔️ `AdministratorAccess`



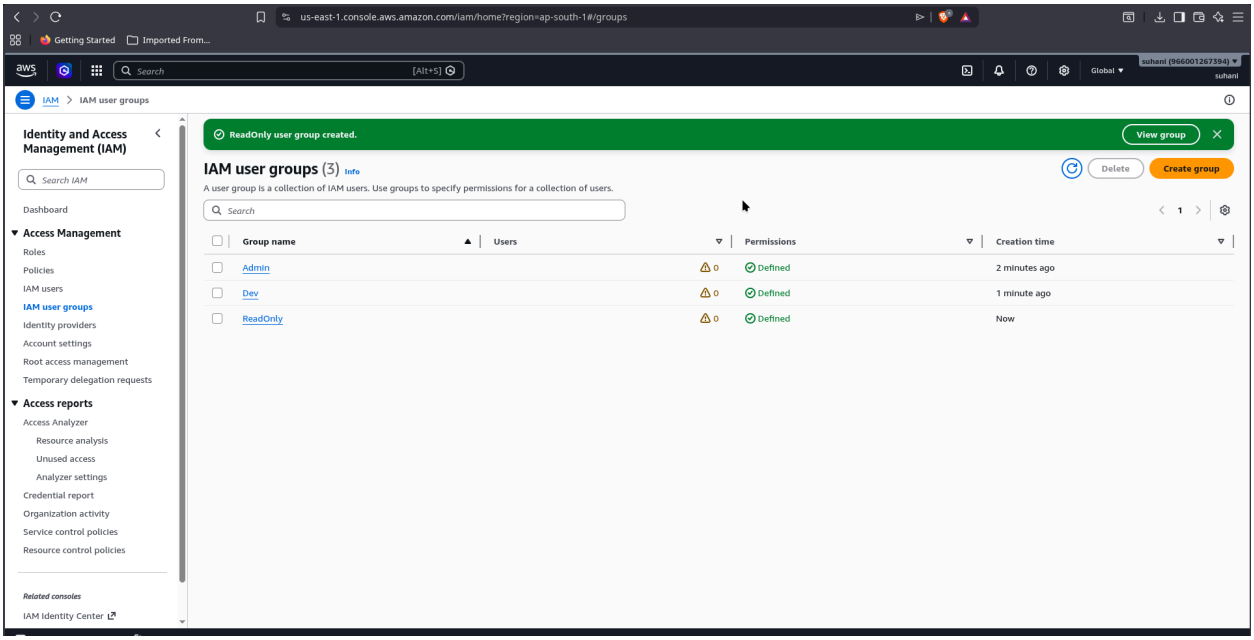

## 2. Dev

- Group name: `Dev`
- Attach:
  - ✔️ `AmazonS3FullAccess`




# 3. ReadOnly

- Group name: `ReadOnly`
- Attach:
  - ✔️ `ReadOnlyAccess`




# Step 2: Apply Least Privilege

👉 Important concept:

- Admin → full access
- Dev → only needed services
- ReadOnly → only viewing

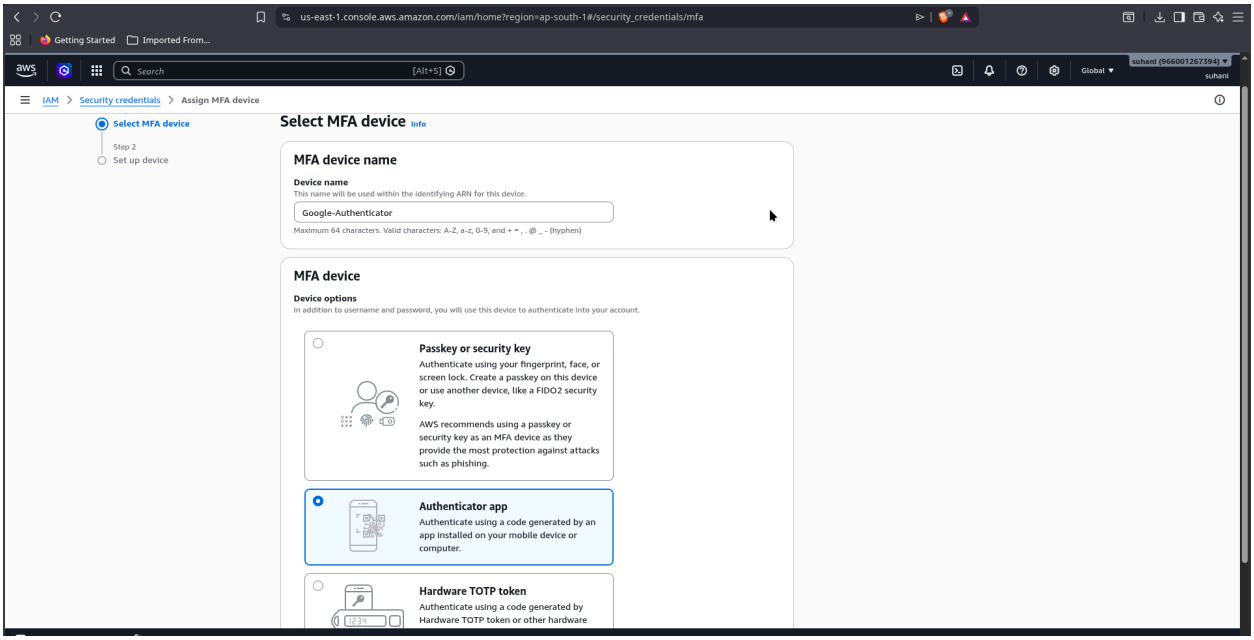
# Step 3: Enable MFA for Root

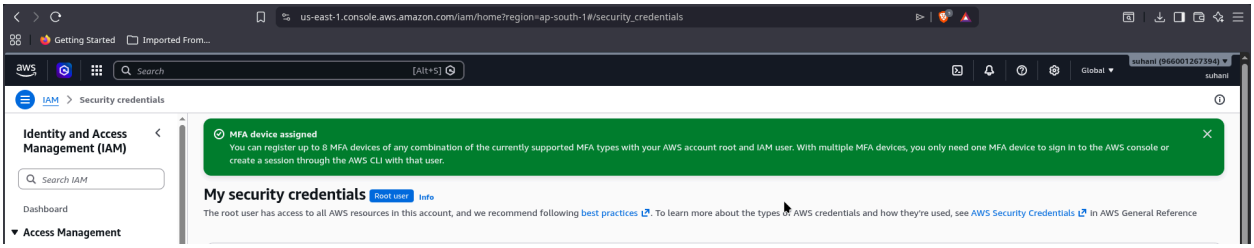
1. Click your account (top right)
2. Go to:
   - **Security credentials**
3. Find:
   - **Multi-Factor Authentication (MFA)**

4. Click **Assign MFA device**

👉 Use:

- Google Authenticator / Authy

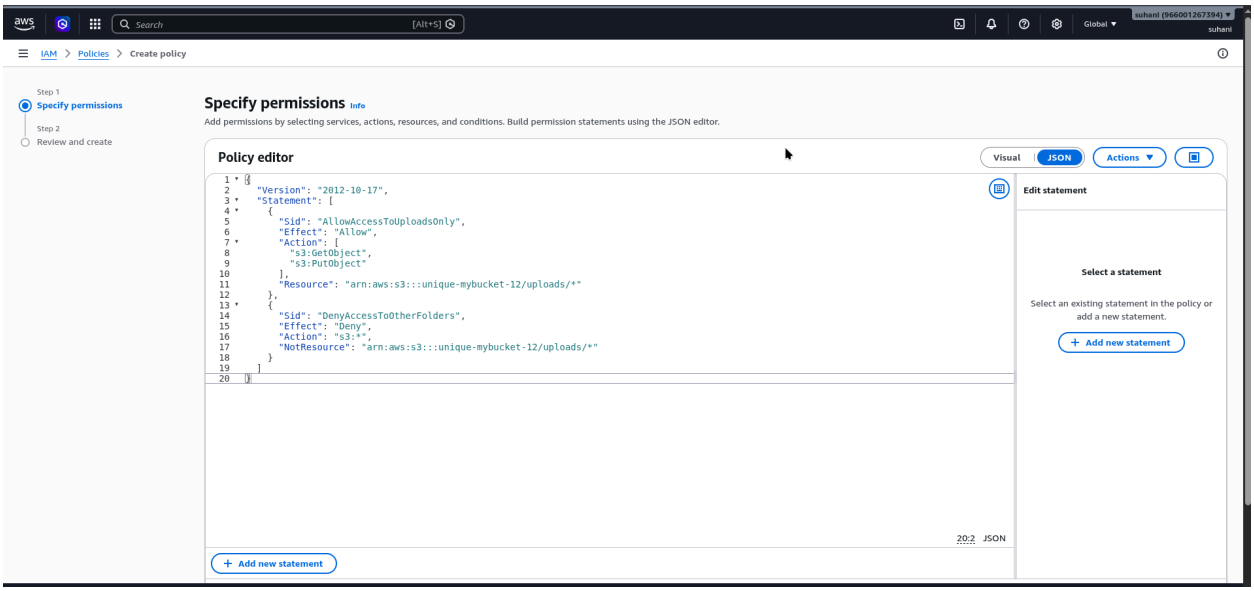


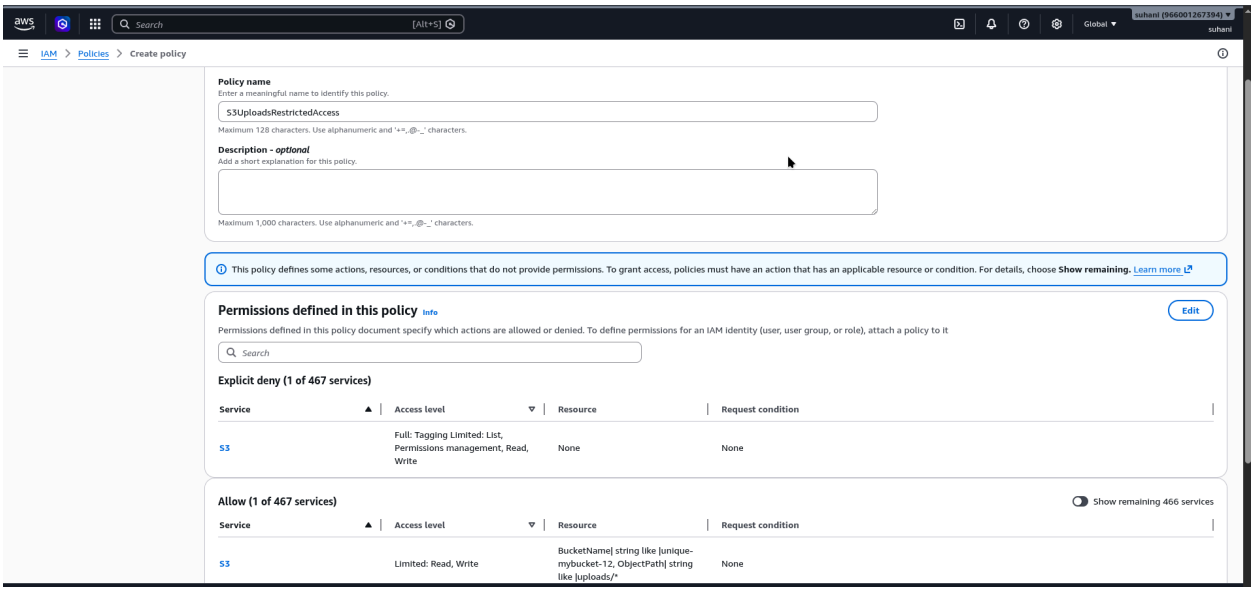



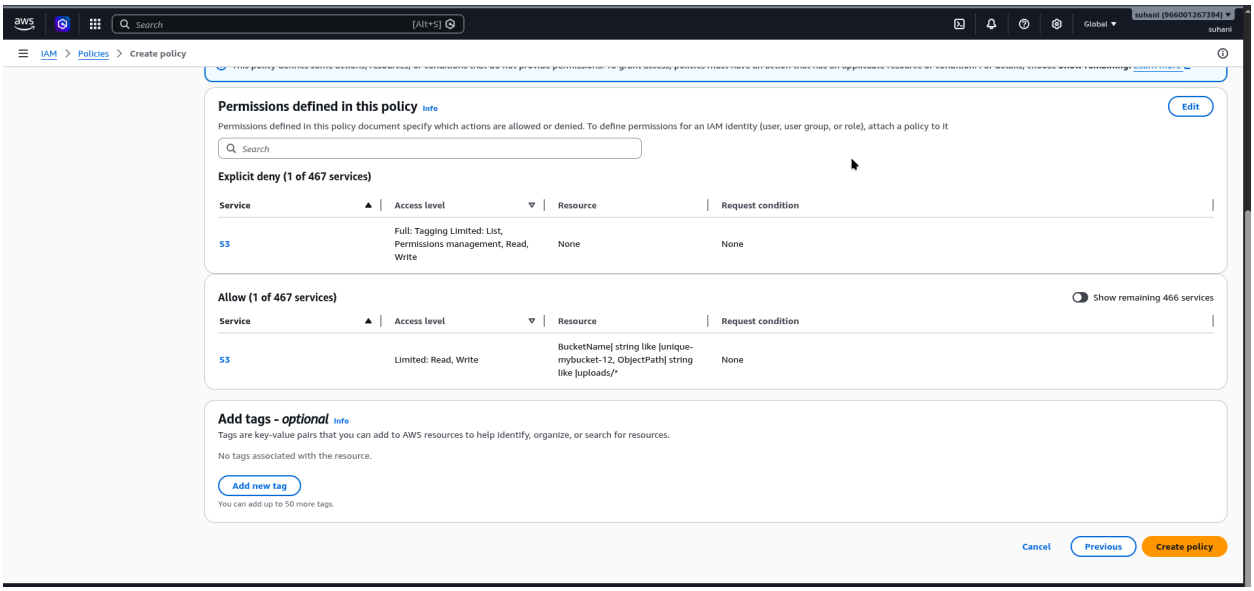

# Step 4: Create Custom Policy (S3 Restricted Folder)

Go to:

- **Policies → Create policy**

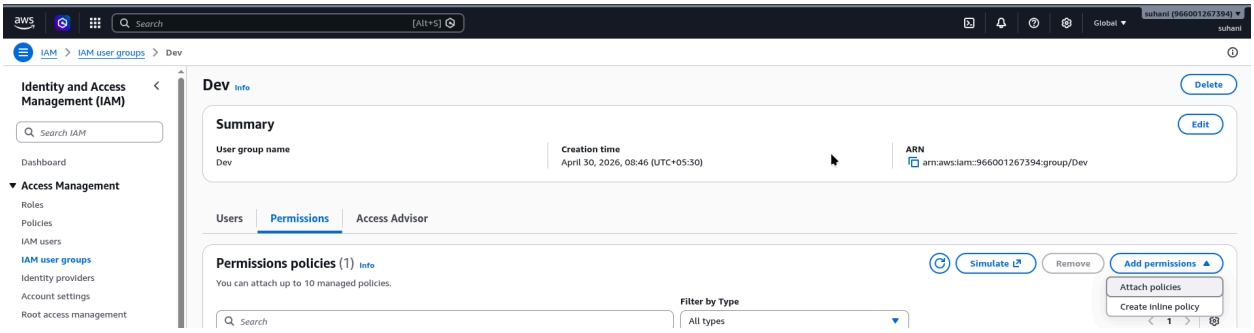

IAM > Policies > Create policy
Step 1
Specify permissions
Step 2
Review and create
Specify permissions Info
Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.
Policy editor
Visual
JSON
Actions
{
"Version": "2012-10-17",
"Statement": [
{
"Sid": "AllowAccessToUploadsOnly",
"Effect": "Allow",
"Action": [
"s3:GetObject",
"s3:PutObject"
],
"Resource": "arn:aws:s3:::unique-mybucket-12/uploads/*"
},
{
"Sid": "DenyAccessToOtherFolders",
"Effect": "Deny",
"Action": "s3:*",
"NotResource": "arn:aws:s3:::unique-mybucket-12/uploads/*"
}
]
}
20:2 JSON
Add new statement
Edit statement
Select a statement
Select an existing statement in the policy or add a new statement.
Add new statement

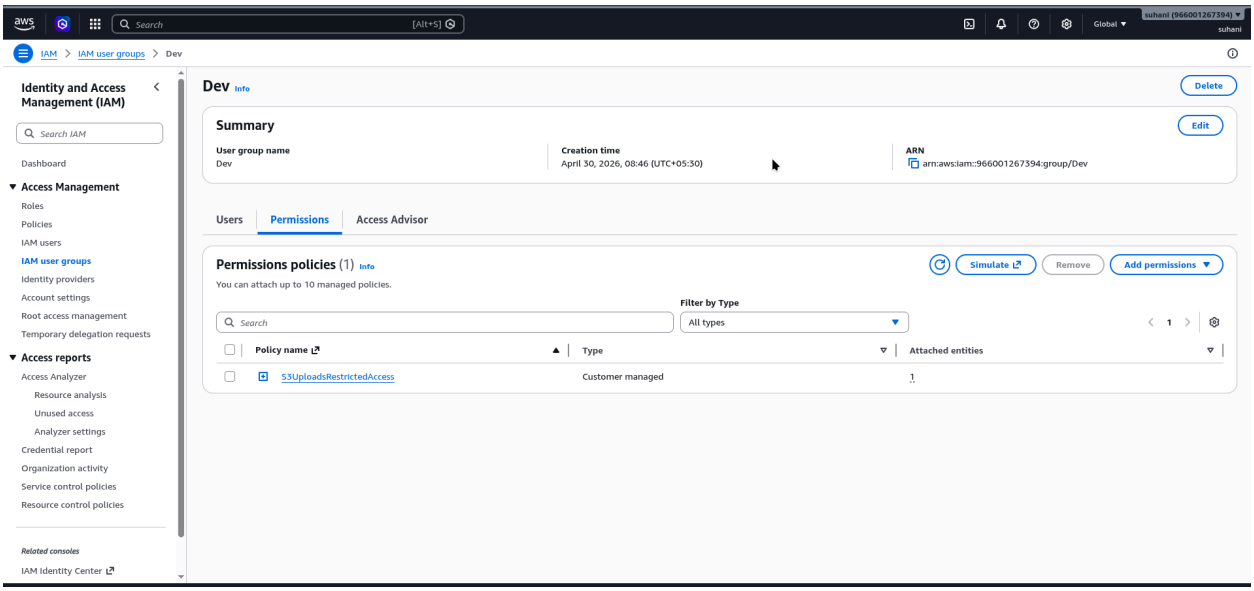

IAM > Policies > Create policy
Policy name
Enter a meaningful name to identify this policy.
S3UploadsRestrictedAccess
Maximum 128 characters. Use alphanumeric and '+=,.@-_' characters.
Description - optional
Add a short explanation for this policy.
Maximum 1,000 characters. Use alphanumeric and '+=,.@-_' characters.
This policy defines some actions, resources, or conditions that do not provide permissions. To grant access, policies must have an action that has an applicable resource or condition. For details, choose Show remaining. Learn more
Permissions defined in this policy Info
Edit
Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for an IAM identity (user, user group, or role), attach a policy to it
Explicit deny (1 of 467 services)
Service
Access level
Resource
Request condition
S3
Full: Tagging Limited: List, Permissions management, Read, Write
None
None
Allow (1 of 467 services)
Show remaining 466 services
Service
Access level
Resource
Request condition
S3
Limited: Read, Write
BucketName| string like |unique-mybucket-12, ObjectPath| string like |uploads/*
None




# Step 5: Attach Policy to Dev Group

- Go to **User groups → Dev**
- Click **Add permissions**
- Attach your custom policy




aws
Search
[Alt+S]
Global
suhani (966001267394)
suhani
IAM > IAM user groups > Dev
Identity and Access Management (IAM)
Search IAM
Dashboard
Access Management
Roles
Policies
IAM users
IAM user groups
Identity providers
Account settings
Root access management
Temporary delegation requests
Access reports
Access Analyzer
Resource analysis
Unused access
Analyzer settings
Credential report
Organization activity
Service control policies
Resource control policies
Related consoles
IAM Identity Center
Dev Info
Delete
Summary
Edit
User group name
Dev
Creation time
April 30, 2026, 08:46 (UTC+05:30)
ARN
arn:aws:iam::966001267394:group/Dev
Users
Permissions
Access Advisor
Permissions policies (1) Info
You can attach up to 10 managed policies.
Simulate
Remove
Add permissions
Filter by Type
Search
All types
1
Policy name
Type
Attached entities
S3UploadsRestrictedAccess
Customer managed
1